

Module 11: Relational Algebra & Calculus

1. Relational Operations Taxonomy

Relational Algebra Operations & Query Languages

Fundamental Operations (σ , π , $\times$, $\cup$, $-$), Joins ($\bowtie$), Division ($\div$) and Calculus

SELECTION: $\sigma_{\{Marks > 80\}}(R)$

Roll	Name	Marks	
101	Akhil	92	✓ SELECTED
102	Rahul	75	(Filtered out)
103	Pooja	88	✓ SELECTED

Horizontal Filtering of Tuples
Degree remains same; Cardinality $\leq$ original

PROJECTION: $\pi_{\{Roll, Name\}}(R)$

Roll	Name	Marks
101	Akhil	
102	Rahul	Dropped
103	Pooja	

Vertical Filtering of Columns
Auto-eliminates duplicate tuples!

JOIN OPERATIONS ($R \bowtie S$)

- Natural Join ($R \bowtie S$):**
Equi-join on common attributes; duplicate column dropped.
- Left Outer Join ($R \bowtie\!\!\!\rhd S$):**
All left tuples kept; unmatched right padded with NULL.
- Right Outer Join ($R \bowtie\!\!\!\lhd S$):**
All right tuples kept; unmatched left padded with NULL.
- Full Outer Join ($R \bowtie\!\!\!\rhd\!\!\!\lhd S$):**
All tuples from both sides kept with NULL padding.

DIVISION OPERATOR ($\div$) & RELATIONAL CALCULUS (TRC / DRC)

Division Operator ($R \div S$) - "FOR ALL" Queries

Used for queries with condition "Find entities associated with ALL..."
Example: "Find students enrolled in ALL courses taught by Prof. Rao"
Formula: $R \div S = \pi_{\{R-S\}}(R) - \pi_{\{R-S\}}((\pi_{\{R-S\}}(R) \times S) - R)$
If $R(A, B)$ has degree 2 and $S(B)$ has degree 1 $\rightarrow (R \div S)$ has degree 1 (A).
Highly popular in GATE and AKTU 10-mark examinations!

Relational Calculus: Declarative Formal Languages

- TRC (Tuple Relational Calculus):** $\{ t \mid P(t) \}$
Quantifiers: $\exists$ (There exists), $\forall$ (For all). Safety rule restricts infinite sets.
- DRC (Domain Relational Calculus):** $\{ \langle x1, x2 \rangle \mid P(x1, x2) \}$
Variables range over attribute domain values instead of whole tuples.
- Codd's Theorem: Relational Algebra $\equiv$ Safe TRC $\equiv$ Safe DRC!**

2. Fundamental vs Derived Operations

Operation	Notation	Type	Core Definition
Selection	$\sigma_p(R)$	Fundamental	Horizontal filtering of tuples
Projection	$\pi_{cols}(R)$	Fundamental	Vertical filtering of columns (auto-deduplication)
Cartesian Product	$R \times S$	Fundamental	Pairwise tuple combinations
Set Union	$R \cup S$	Fundamental	Combines union-compatible tables
Set Difference	$R - S$	Fundamental	Tuples in R but not in S
Natural Join	$R \bowtie S$	Derived	Equi-join on shared attributes
Division	$R \div S$	Derived	Answers "ALL" universally quantified queries

3. Codd's Theorem

Relational Completeness: Relational Algebra $\equiv$ Safe Tuple Relational Calculus (TRC) $\equiv$ Safe Domain Relational Calculus (DRC). SQL is built on top of these foundations.